# DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
# UNIVERSITY OF BARISAL

**Mid-term – 17 Sep, 2025 (Course Code: CSE-3203)**

**Course Title: Theory of Computation**

3rd Year 2nd Semester, Session: 2020-21

**Time:** 50+50 Minutes, **Marks:** 40 (will be converted to 20% marks)

| NAME: | ROLL: |
|---|---|

**Answer ALL Questions**

**Question 1: [Marks 4]**

Make an NFA: Strings where the first symbol is present somewhere later on at least once.

**Question 2: [Marks 8]**

i) Analyze about the "Bad Case for the Subset Construction" during DFA to NFA conversion.

ii) Describe the "pigeonhole principle" regarding the subset construction.

**Question 3: [Marks 4]**

*Build* an ε-NFA for the following language:

$$L = \{w \mid w \text{ is empty, or if non-empty will end in } 01\}$$

**Question 4: [Marks 8]**

i) Prove that if we add a finite set of strings to a regular language, the result is a regular language.

ii) A language L is accepted by some ε-NFA if and only if L is accepted by some DFA.

**Question 5: [Marks 4]**

Convert the following RE to ε-NFA: (0+1)*01(0+1)*

**Question 6: [Marks 8]**

i) Prove the theorem "a mod b = b mod a , if and only if a = b .

ii) By method of induction, proof the theorem "if $x \geq 4$ then $2^x \geq x^2$ .

**Question 7: [Marks 4]**

Describe the importance of Automata Theory.

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